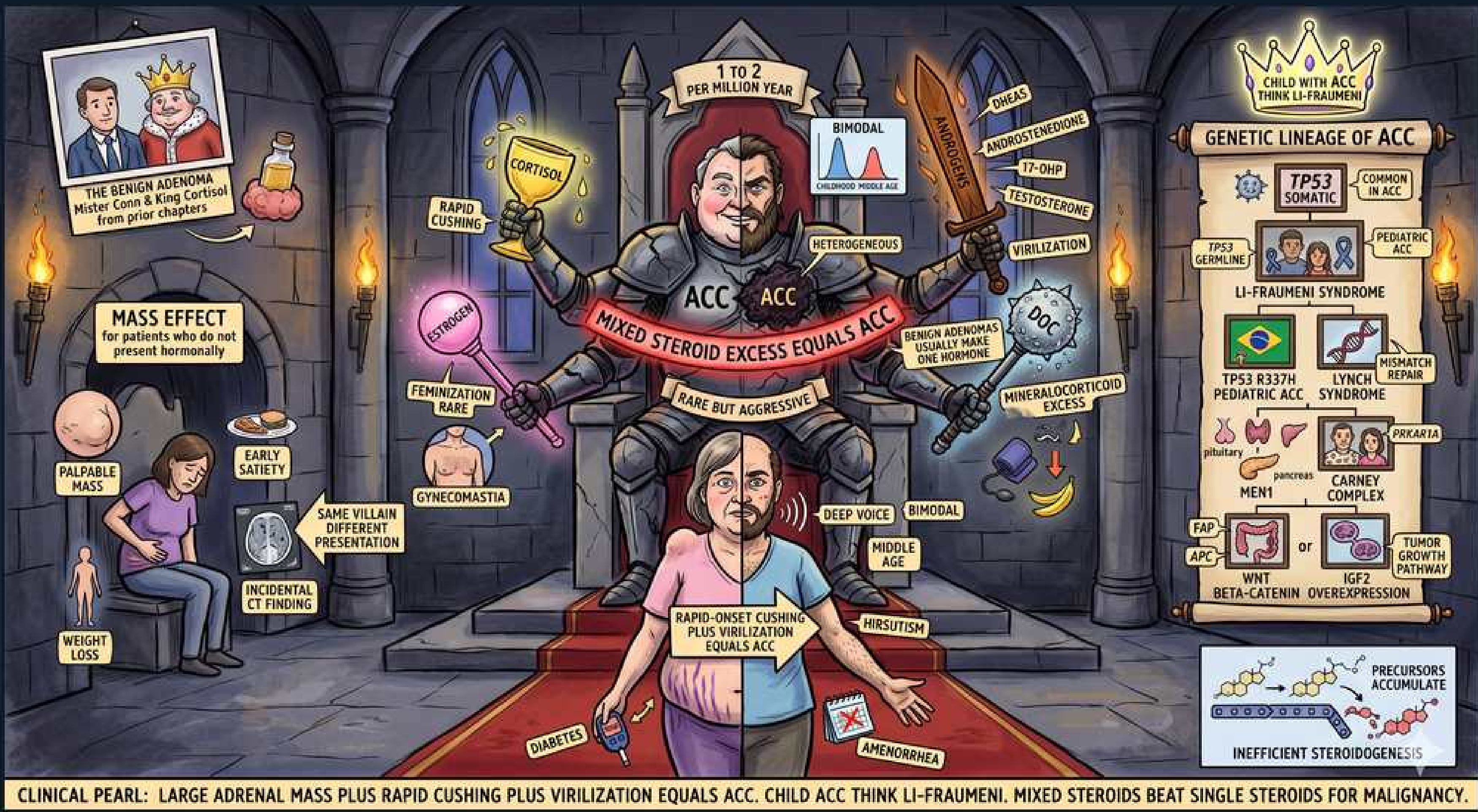
Adrenal — ACC Villain Reveal



1. ACC villain

WHAT YOU SEE

Armored multi-armed villain labeled 'ACC'

WHAT IT ENCODES

Adrenocortical carcinoma — a rare, aggressive malignancy of the adrenal CORTEX (not medulla; that's pheochromocytoma). The multiple arms each holding a different steroid capture its hallmark: a disorganized tumor that pumps out several steroid lines at once because it has lost the normal zona-specific enzyme control. Most are functional (~60%) and present with hormone excess; the rest present as mass effect. 5-year survival is poor (~35-50% overall, <15% if metastatic) and complete en-bloc surgical resection by an experienced center is the only curative option.

BOARD TRAP

Don't confuse cortex (ACC, steroids) with medulla (pheochromocytoma, catecholamines). And a functioning adrenal mass is NOT automatically benign — size, growth rate, and mixed hormone output are what point to malignancy.

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2. 1-2 per million / year

WHAT YOU SEE

Banner '1 TO 2 PER MILLION YEAR', bimodal age graph

WHAT IT ENCODES

ACC incidence is ~1-2 per million per year, making it one of the rarest endocrine cancers. The age distribution is BIMODAL: a first peak in young children (<5 yr, where germline TP53/Li-Fraumeni dominates) and a second, larger peak in adults aged 40-50. Slight female predominance. Rarity is exactly why a confirmed ACC on a board question is usually accompanied by a giveaway clue (very large mass, rapid virilizing Cushing, or a child).

BOARD TRAP

Because it is so rare, the board will not expect you to diagnose ACC from subtle findings — it will hand you the classic large-mass + mixed-steroid + rapid-onset picture.

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3. Mixed steroid excess = ACC

WHAT YOU SEE

Red belt banner 'MIXED STEROID EXCESS EQUALS ACC', 'RARE BUT AGGRESSIVE'

WHAT IT ENCODES

The single most useful board discriminator: CO-SECRETION of multiple steroid classes (cortisol + androgens ± mineralocorticoids ± estrogen) strongly favors ACC over a benign adenoma. Benign adenomas are typically monoclonal and secrete ONE hormone (usually just cortisol, or just aldosterone). ACC's loss of differentiation means inefficient, promiscuous steroidogenesis — note the 'precursors accumulate / inefficient steroidogenesis' panel. Mnemonic: 'Many arms, many steroids = malignant.'

BOARD TRAP

A single isolated hormone excess (pure cortisol, or pure aldosterone) favors a benign adenoma — do NOT jump to ACC. It is the COMBINATION of cortisol + androgen excess (rapid Cushing with virilization) that flags carcinoma.

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4. Rapid Cushing

WHAT YOU SEE

'RAPID CUSHING', cortisol vial

WHAT IT ENCODES

Rapid-onset hypercortisolism (Cushing developing over weeks-to-few-months rather than years) is a red flag for malignancy — either ACC or ectopic ACTH (e.g., small-cell lung cancer). The speed reflects an aggressive tumor's high cortisol output. Contrast with pituitary Cushing disease, which usually evolves slowly over years with the classic indolent body-habitus change.

BOARD TRAP

Slowly progressive Cushing over years points to a benign pituitary adenoma (Cushing disease); RAPID onset shifts you toward ACC or ectopic ACTH instead.

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5. Virilization

WHAT YOU SEE

'VIRILIZATION', 17-OHP, DHEA-S/testosterone, deep voice

WHAT IT ENCODES

Virilization in a woman — hirsutism, deep voice, clitoromegaly, acne, male-pattern balding, oligo/amenorrhea — driven by tumor androgens (DHEA-S, testosterone, 17-OH-progesterone). A markedly elevated DHEA-S (an adrenal-specific androgen) points to an adrenal source and, when combined with cortisol excess, to ACC. In a woman, new androgen excess plus a large adrenal mass is ACC until proven otherwise.

BOARD TRAP

Mild hirsutism with normal/mildly elevated androgens is usually PCOS or idiopathic; it is the RAPID, severe virilization with high DHEA-S and a mass that signals ACC. PCOS does not cause a large adrenal mass.

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6. Feminization / gynecomastia

WHAT YOU SEE

'FEMINIZATION', 'GYNECOMASTIA'

WHAT IT ENCODES

Estrogen-secreting ACC in a MAN causes feminization: gynecomastia, decreased libido, testicular atrophy. This is rare but, like virilization in a woman, is a near-specific tumor marker — benign adenomas essentially never feminize. The tumor's aromatase activity or estrogen-precursor output drives it.

BOARD TRAP

Gynecomastia has many benign causes (drugs, cirrhosis, hypogonadism, hyperthyroidism), but feminization PLUS an adrenal mass is a feminizing ACC and warrants urgent surgical workup, not reassurance.

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7. Mass effect / palpable mass

WHAT YOU SEE

Left: 'PALPABLE MASS', 'EARLY SATIETY', 'WEIGHT LOSS', flank

WHAT IT ENCODES

The non-hormonal (~40%) presentation: a large abdominal/flank mass causing local symptoms — palpable mass, abdominal fullness, early satiety, flank/back pain, and constitutional weight loss. ACCs are typically large at diagnosis (often >6 cm). Any adrenal mass >4 cm carries meaningful malignancy risk; >6 cm should usually be resected regardless of imaging phenotype.

BOARD TRAP

A small (<4 cm), homogeneous, lipid-rich (<10 HU on unenhanced CT) incidentaloma is almost always a benign adenoma — don't over-call ACC. Size >4-6 cm and weight loss are what raise the malignancy concern.

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8. Incidental CT finding

WHAT YOU SEE

'INCIDENTAL CT FINDING', 'SAME VILLAIN DIFFERENT PRESENTATION'

WHAT IT ENCODES

Some ACCs are found incidentally when imaging is done for another reason. On CT, malignant features are: size >4 cm, unenhanced attenuation >10-20 HU (lipid-poor), heterogeneity/necrosis/calcification, irregular margins, and slow contrast washout (absolute washout <60%). These imaging red flags overlap with the same villain seen via hormones — 'same villain, different presentation.'

BOARD TRAP

A benign adenoma is small, smooth, lipid-rich (<10 HU) with rapid contrast washout (>60%). High HU + large size + delayed washout = worry about ACC or metastasis, and prompts biochemical workup before any biopsy.

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9. Rapid Cushing + virilization = ACC

WHAT YOU SEE

Banner 'RAPID-ONSET CUSHING PLUS VIRILIZATION EQUALS ACC'

WHAT IT ENCODES

The signature board vignette: a woman with rapidly progressive Cushingoid features AND new virilization (or a man with Cushing + feminization). This cortisol + androgen co-secretion is the textbook ACC presentation and is the highest-yield single line of this scene. It directly contrasts with the slow, cortisol-only picture of pituitary Cushing disease.

BOARD TRAP

If the stem gives Cushing alone (no androgen/estrogen excess, slow onset, small lesion), think benign adenoma or pituitary disease — adding rapid virilization is what converts the answer to ACC.

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10. Li-Fraumeni / TP53

WHAT YOU SEE

Right panel 'TP53 SOMATIC/GERMLINE', 'LI-FRAUMENI', Lynch, MEN1, Carney, Beckwith-Wiedemann

WHAT IT ENCODES

Genetics of ACC: TP53 is the key driver — somatic TP53 mutations are common in sporadic adult ACC, and GERMLINE TP53 defines Li-Fraumeni syndrome (also breast cancer, sarcoma, brain tumors, leukemia). Other syndromic associations shown: Lynch (MMR), MEN1, Carney complex, Beckwith-Wiedemann (IGF2 overexpression), and WNT/β-catenin pathway activation. IGF2 overexpression is the most frequent molecular abnormality in adult ACC.

BOARD TRAP

Don't limit Li-Fraumeni to ACC alone — the germline TP53 carrier is at risk for the full spectrum (sarcoma, premenopausal breast cancer, brain tumor, leukemia/'SBLA'), which is what the board tests around a pediatric ACC.

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11. Child ACC → Li-Fraumeni

WHAT YOU SEE

Crowned child 'CHILD WITH ACC THINK LI-FRAUMENI'

WHAT IT ENCODES

Pediatric ACC is strongly tied to germline TP53. In southern Brazil (note the Brazilian flag in the scene) a founder TP53 R337H mutation makes childhood ACC strikingly common — hence the country flag as a mnemonic anchor. Any child presenting with ACC should trigger germline TP53 testing and genetic counseling for the family.

BOARD TRAP

A child with virilization (precocious puberty signs) plus an adrenal mass is ACC/Li-Fraumeni territory — do not dismiss it as benign congenital adrenal hyperplasia without imaging; CAH does not produce a discrete large mass.

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12. Mineralocorticoid excess

WHAT YOU SEE

'MINERALOCORTICOID EXCESS' label with arm/banner on right side of villain

WHAT IT ENCODES

Some ACCs also oversecrete mineralocorticoids (aldosterone or its precursors), producing hypertension with hypokalemic metabolic alkalosis. Combined with the very high cortisol (which overwhelms renal 11β-HSD2 and spills onto the mineralocorticoid receptor), ACC patients can have severe, refractory HTN and profound hypokalemia — another facet of the 'mixed steroid' fingerprint.

BOARD TRAP

Isolated aldosterone excess with a small adenoma is Conn syndrome (benign), not ACC — it's the aldosterone seen ALONGSIDE cortisol/androgen excess and a large mass that points to carcinoma.

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13. Clinical pearl

WHAT YOU SEE

'LARGE ADRENAL MASS PLUS RAPID CUSHING PLUS VIRILIZATION EQUALS ACC. CHILD ACC THINK LI-FRAUMENI. MIXED STEROIDS BEAT SINGLE STEROIDS FOR MALIGNANCY'

WHAT IT ENCODES

Three locked take-home rules: (1) Large mass (>4-6 cm) + rapid Cushing + virilization = ACC. (2) Any child with ACC → suspect germline TP53 / Li-Fraumeni. (3) Mixed/multiple steroid excess beats single-steroid excess for malignancy — benign adenomas secrete one hormone, carcinomas secrete many. Management is complete surgical resection ± mitotane (an adrenolytic) ± etoposide-doxorubicin-cisplatin chemotherapy for advanced disease.

BOARD TRAP

The seductive distractors are 'benign adenoma' (single hormone, small, lipid-rich) and 'pituitary Cushing disease' (slow, cortisol-only) — the rapidity, the mass size, and the steroid MIXTURE are the tie-breakers that select ACC.
